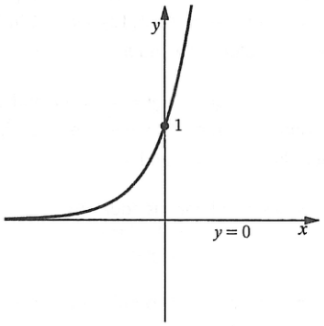
Graph Sketching: Exponentials and Logarithms

Questions 1–26

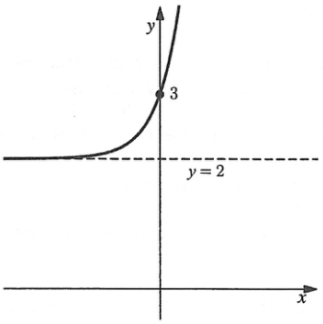
Question 1. Find the equation of each of the following graphs in the form

$$y = 2^x + b$$

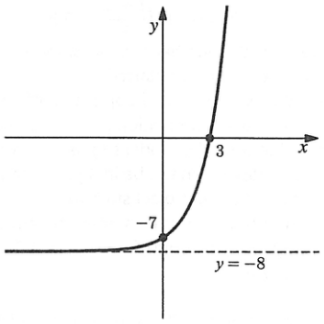
a)



b)



c)



Question 2. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

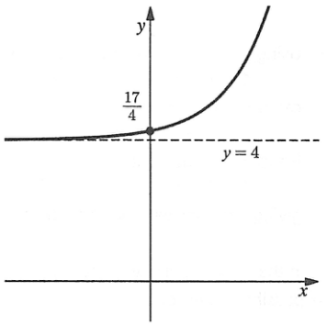
a) $y = 3^x - 9$

b) $y = 4^x + 6$

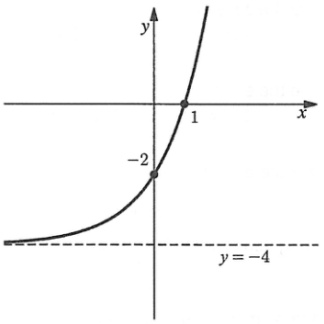
Question 3. Find the equation of each of the following graphs in the form

$$y = 2^{x-a} + b$$

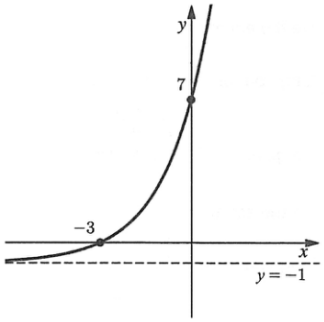
a)



b)



c)



Question 4. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

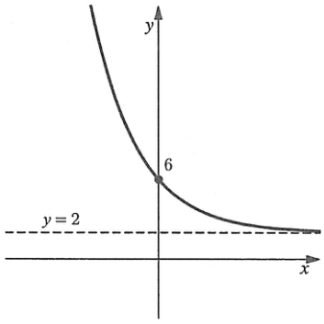
a) $y = 2^{x-4} - 4$

b) $y = 3^{x+2} + 2$

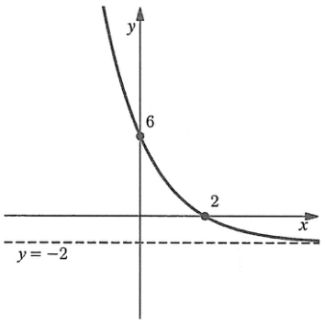
Question 5. Find the equation of each of the following graphs in the form

$$y = 2^{a-x} + b$$

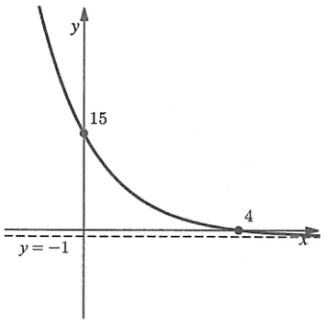
a)



b)



c)



Question 6. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

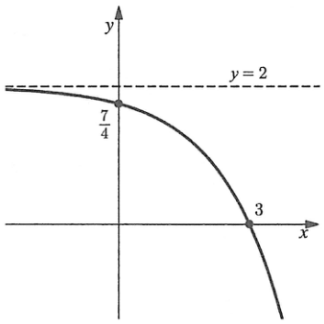
a) $y = 3^{2-x} + 4$

b) $y = 5^{-1-x} - 1$

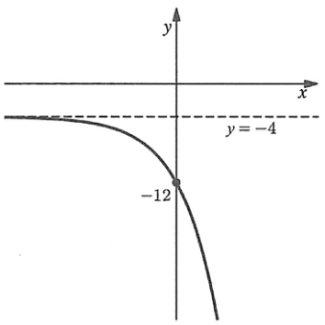
Question 7. Find the equation of each of the following graphs in the form

$$y = b - 2^{x-a}$$

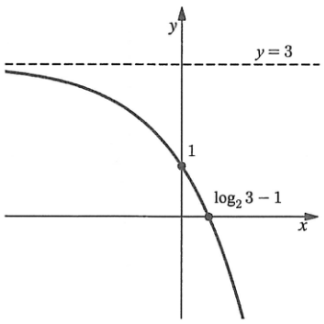
a)



b)



c)



Question 8. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

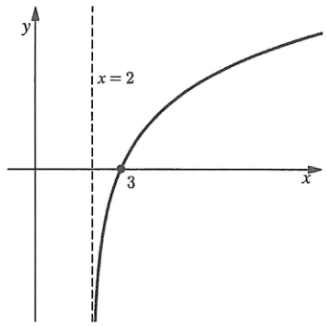
a) $y = 2 - 3^{x-1}$

b) $y = 3 - 2^{1-x}$

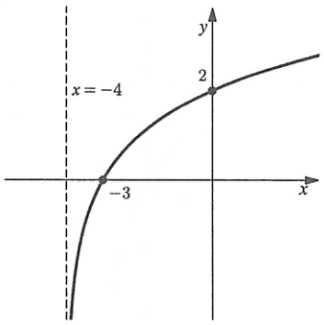
Question 9. Find the equation of each of the following graphs in the form

$$y = \log_2(x - a)$$

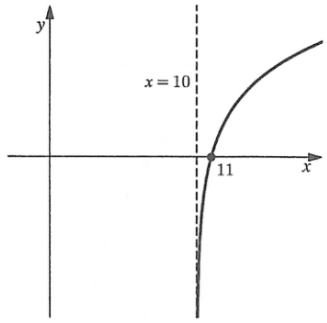
a)



b)



c)



Question 10. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

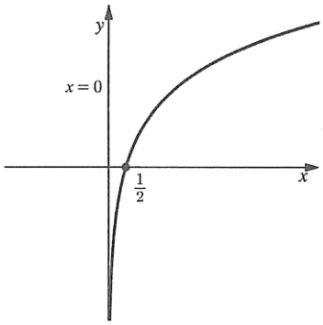
a) $y = \log_3(x - 2)$

b) $y = \log_5(x + 4)$

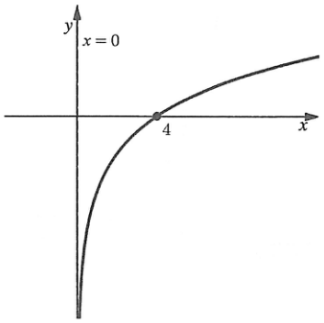
Question 11. Find the equation of each of the following graphs in the form

$$y = \log_2 x + b$$

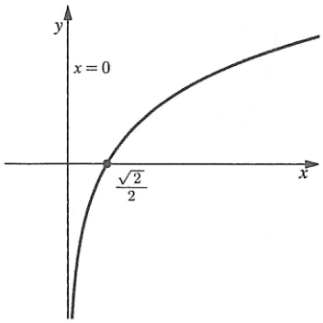
a)



b)



c)



Question 12. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

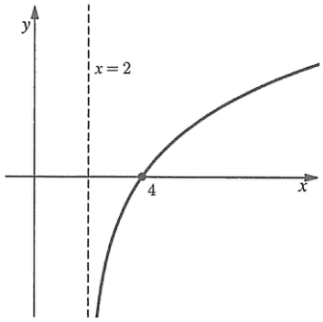
a) $y = \log_5 x - 5$

b) $y = \log_3 x + 4$

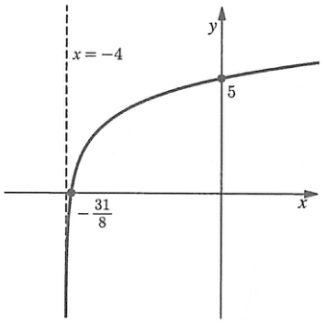
Question 13. Find the equation of each of the following graphs in the form

$$y = \log_2(x - a) + b$$

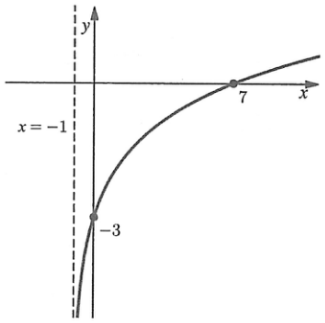
a)



b)



c)



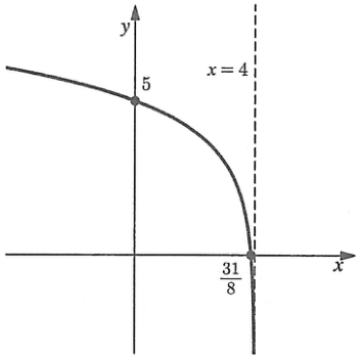
Question 14. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_4(x - 2) + 1$

b) $y = \log_3(x + 3) - 3$

Question 15. Find the equation of the following graph in the form

$$y = \log_2(a - x) + b$$

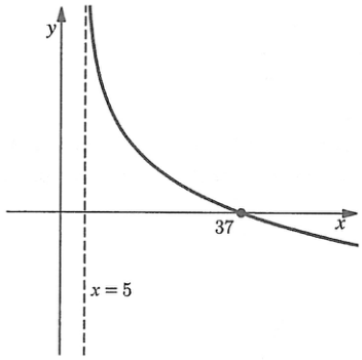


Question 16. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_4(5 - x) + 3$
b) $y = \log_5(10 - x) - 2$
c) $y = \log_3(-3 - x) + 1$
d) $y = \log_4(-8 - x) - 2$

Question 17. Find the equation of the following graph in the form

$$y = b - \log_2(x - a)$$



Question 18. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 3 - \log_3(x - 3)$
b) $y = 2 - \log_4(x + 2)$
c) $y = 1 - \log_5(3 - x)$
d) $y = 2 - \log_3(9 - x)$

Question 19. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = 2^x$ and $y = 4^x$

b) $y = 3^x$ and $y = 4^{-x}$

c) $y = 2^{-x}$ and $y = 3^{-x}$

Question 20. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -5^x$ and $y = -3^x$

b) $y = -2^x$ and $y = 3^x$

c) $y = -4^{-x}$ and $y = -2^{-x}$

Question 21. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = \log_2 x$ and $y = \log_4 x$

b) $y = \log_5 x$ and $y = \log_3(-x)$

c) $y = \log_6(-x)$ and $y = \log_2(-x)$

Question 22. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -\log_4 x$ and $y = -\log_3 x$

b) $y = -\log_8 x$ and $y = \log_4 x$

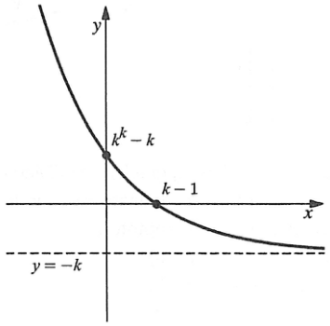
c) $y = -\log_2(-x)$ and $y = -\log_4(-x)$

Question 23. It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

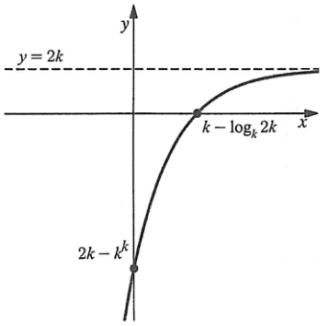
a) $y = k^x$
b) $y = k^x - k$
c) $y = k^{x-k} + k$

Question 24. It is given that $k > 1$. Write down a suitable equation for each of the following curves.

a)



b)



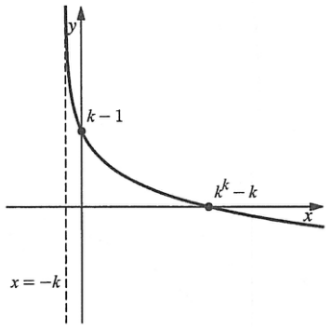
Question 25. It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_k x + k$

b) $y = \log_k(x + k) - k$

Question 26. It is given that $k > 1$. Write down a suitable equation for each of the following curves.

a)



b)

